

9.1.5 Connection between Hypothesis Tests and Confidence Intervals

So far the result of a test has been determined through a:

- **critical region Z**: if the value of the test statistic determined by the sample is in the critical region, we will reject the null hypothesis.
- **P-value**: if the P -value determined by the sample is less than or equal to α , we will reject the null hypothesis.

In the case of the 25 measurements form a normal distribution with known $\sigma = 15$ and $\bar{x} = 292$, the unknown parameter μ could be estimated with a **95% confidence interval**:

$$[\vartheta_1, \vartheta_2] = \left[\bar{x} - z_{1/2\alpha} \cdot \frac{\sigma}{\sqrt{n}}; \bar{x} + z_{1/2\alpha} \cdot \frac{\sigma}{\sqrt{n}} \right] = [286.12; 297.88]$$

In other words, in 95% of the cases this procedure leads to an interval to which the unknown μ belongs. Because $\mu_0 = 300 \notin [\vartheta_1, \vartheta_2]$, it is unlikely that H_0 is true. H_0 is therefore rejected.

In general: $H_0: \vartheta = \vartheta_0$
 $H_1: \vartheta \neq \vartheta_0$
 $\alpha = \dots$
If $\vartheta_0 \in 100(1 - \alpha)\% \text{ CI } [\vartheta_1; \vartheta_2]$, then H_0 is not rejected

In the case of a **left-sided test**, check that:

$$\mu = 300 \notin < \leftarrow; \bar{x} + z_{\alpha} \cdot \frac{\sigma}{\sqrt{n}} \right] = < \leftarrow; 296.9346]$$

We therefore reject H_0 .

9.1.6 Procedure for Hypothesis Tests

The following steps are distinguished when performing the test procedure:

1. State the hypotheses H_0 and H_1 .
2. Choose α .
3. Determine an appropriate test statistic. Which distribution does the test statistic follow? What value determined by the sample does the test statistic have?
4. Determine a critical region, a P -value or a confidence interval.
5. Draw conclusions, both statistically and in clear terms.

9.3.1 Tests on the Mean of a Normal Distribution (*Variance Unknown*)

If the observations are approximately **normally distributed** with **unknown σ** , the **t distribution** is used:

$$T = \frac{\bar{X} - \mu}{\frac{S}{\sqrt{n}}} \sim t(df = n - 1)$$

Example 2

At a certain point the sale of spinach is stopped because this spinach contained too many radioactive units. The sale can be resumed if the average number of radioactive units falls below 50.

We assume the presence of a normal distribution. Which decision should you make based on the following sample? Choose $\alpha = 0.05$

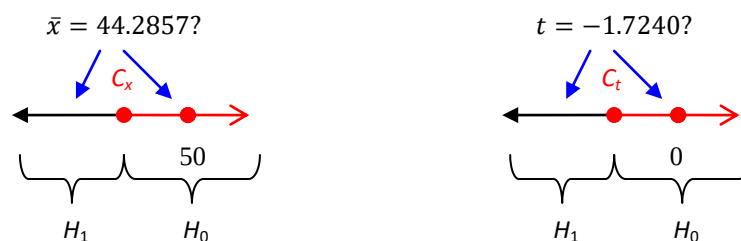
45 55 51 35 52 39 33

Solution

1. $H_0: \mu \geq 50$
 $H_1: \mu < 50$
2. $\alpha = 0.05$
3. Test statistic = $T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t(n - 1) = t(6)$, provided that H_0 is true.

It applies: $t = \frac{44.2857 - 50}{\frac{8.7695}{\sqrt{7}}} = -1.7240$

[gc/stat/edit/1](#) and [gc/stat/calc/1](#)



4. $P(H_1 | H_0) \leq \alpha \Leftrightarrow P(\bar{X} < C_x | \mu = 50) \leq 0.05$
 $\Leftrightarrow P\left(T = \frac{\bar{X} - \mu}{S/\sqrt{n}} < C_t = \frac{g_{\alpha} - \mu}{s/\sqrt{n}} \mid \mu = 50\right) \leq 0.05$
 $\Leftrightarrow C_t = -1.943$

Table V

So $Z = < \leftarrow ; -1.943 >$

5. Since $t = -1.7240 \notin Z$, we do not reject H_0 . The sale of spinach may therefore not resumed.

Remarks

1. $t_0 = -1.7240$ is called the **t-waarde** of the test.
2. $C_t = -1.943$ is called the **critical t-value** of the test.
3. The critical value C_x is determined as follows:

$$-1.943 = C_t = \frac{C_x - 50}{\frac{8.7695}{\sqrt{7}}} \Leftrightarrow C_x = 50 - 1.943 \frac{8.7695}{\sqrt{7}} = 43.5598$$

Since $\bar{x} = 44.2857 \geq 43.5598$, we do not reject H_0 .

4. The **P-value** of the test is determined as follows:

$$\begin{aligned} P(\bar{X} \leq 44.2857 \mid \mu = 50) &= P(T \leq -1.7240) \\ &= \text{tcdf}(-10^{10}, -1.7240, 6) \\ &= 0.0677 \end{aligned} \quad \text{gc/distr/6}$$

Since $p = 0.0677 > \alpha = 0.05$, we do not reject H_0 .

5. With the option **T-Test** of the **graphical calculator**: $p = .0677378744$ gc/stat/tests/2
6. With an **one-sided confidence interval/upper-confidence bound** for μ :

$$\mu_0 = 50 \in < \leftarrow ; \bar{x} + t_{\alpha; n-1} \cdot \frac{s}{\sqrt{n}} = 50.727]$$

So do not reject H_0 .

Remark

It is also possible to determine the upper-confidence bound with the option **T Interval** of the graphical calculator. Choose .9 as **confidence level** and ignore the lower-confidence bound.